

EE-312L Signals and Systems Lab

Laboratory Session 02

Introduction to Plotting in MATLAB

October 3, 2024

Outlines:

1. Objective
2. Plotting in MATLAB
3. Plot commands and subplot

1 Objective

The objective of this lab session is to introduce Plotting in MATLAB using various examples.

2 Plotting in MATLAB

MATLAB is a very reliable and power full tool for plotting. A graph is constructed as a set of points in two or three dimensions. These points are typically connected with a solid line. Commonly a graph of a function is required. However in MATLAB a plot is done using the vectors and matrices not functions.

2.1 How to plot?

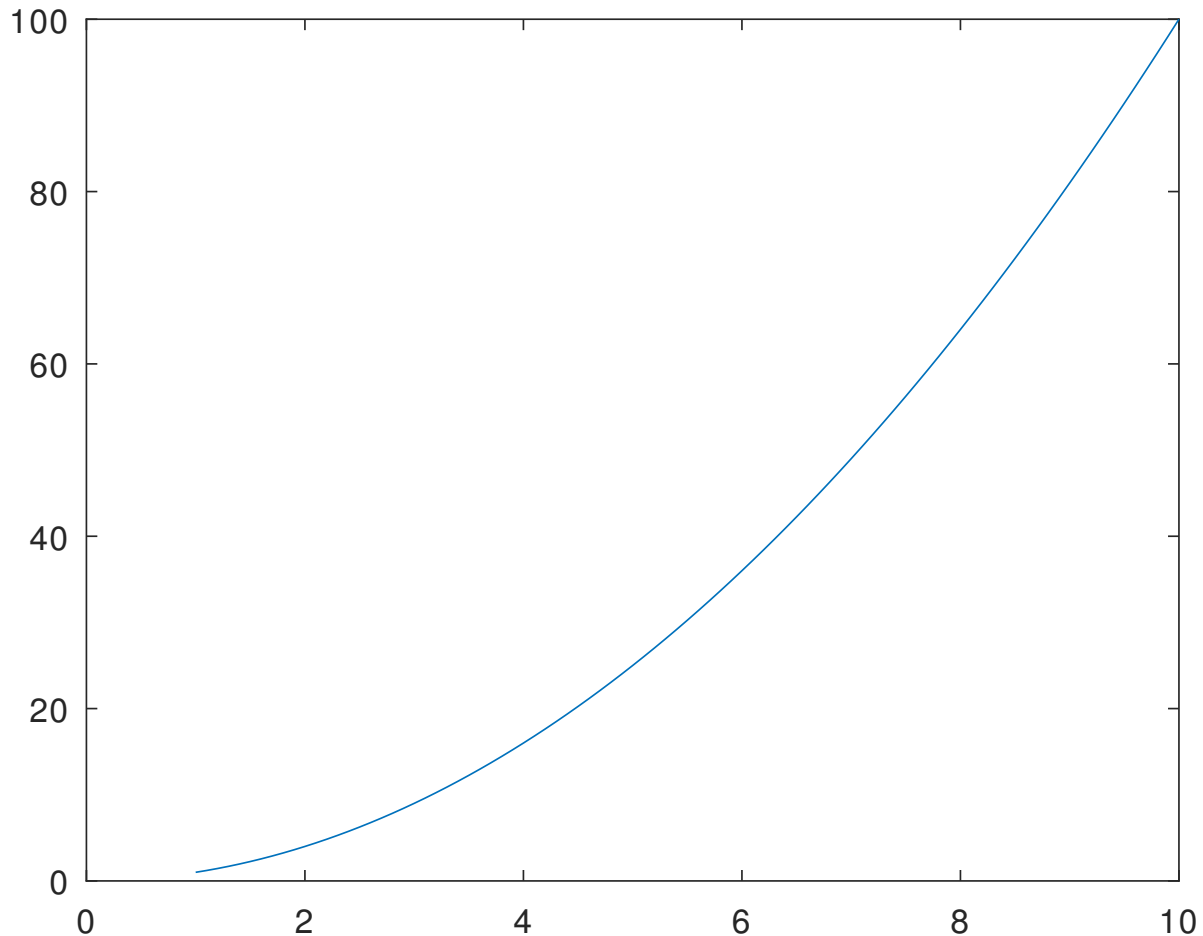
Suppose that we want to plot a function , where is an independent variable. The procedure to plot is as follows:

- i Define an independent with proper range and and spacing, for example: $x = 0 : 0.1 : 5$, where 0 is the initial value, 5 is is the final value and 0.1 is the difference between each two successive values.
- ii Define the function that depends on the independent variable, for example $y = \sin(x)$.
- iii Finally plot the function using command `plot(x,y)`.

2.2 Example: Plotting in Matlab

Plotting in Matlab is done using `plot()` command. Let plot v vs t , where t is independent and V is dependent variable.

```
t=1:1:10;  
v= t.^2;  
plot(t, v)
```



2.3 linspace

You can also use `linspace`, 'linspace' creates an array of samples with linear spacing. The form of `linspace` is

```
y = linspace(start,end);  
y = linspace(start,end,n);
```

```
>> linspace(1,10,10)
ans =
1 2 3 4 5 6 7 8 9 10
```

where start is the starting value, end is the ending value, and n is the number of points to produce in the array. If only the start and end values are specified, linspace produces 100 equally spaced points starting at start and ending at end. For example, we can create an array of 10 evenly spaced points on a linear scale with the command

3 Plotting Several Function in One Figure

It is possible to plot more than one function in the name of figure by employing a different syntax of command Plot.

3.1 Example: Two Sinusoides

```
1 x = -pi:pi/20:pi;
2 y1 = sin(x);
3 y2 = cos(x);
4 plot(x,y1, 'b-');
5 hold on;
6 plot(x,y2, 'k--');
7 hold off;
8 legend('sin x', 'cos x');
```

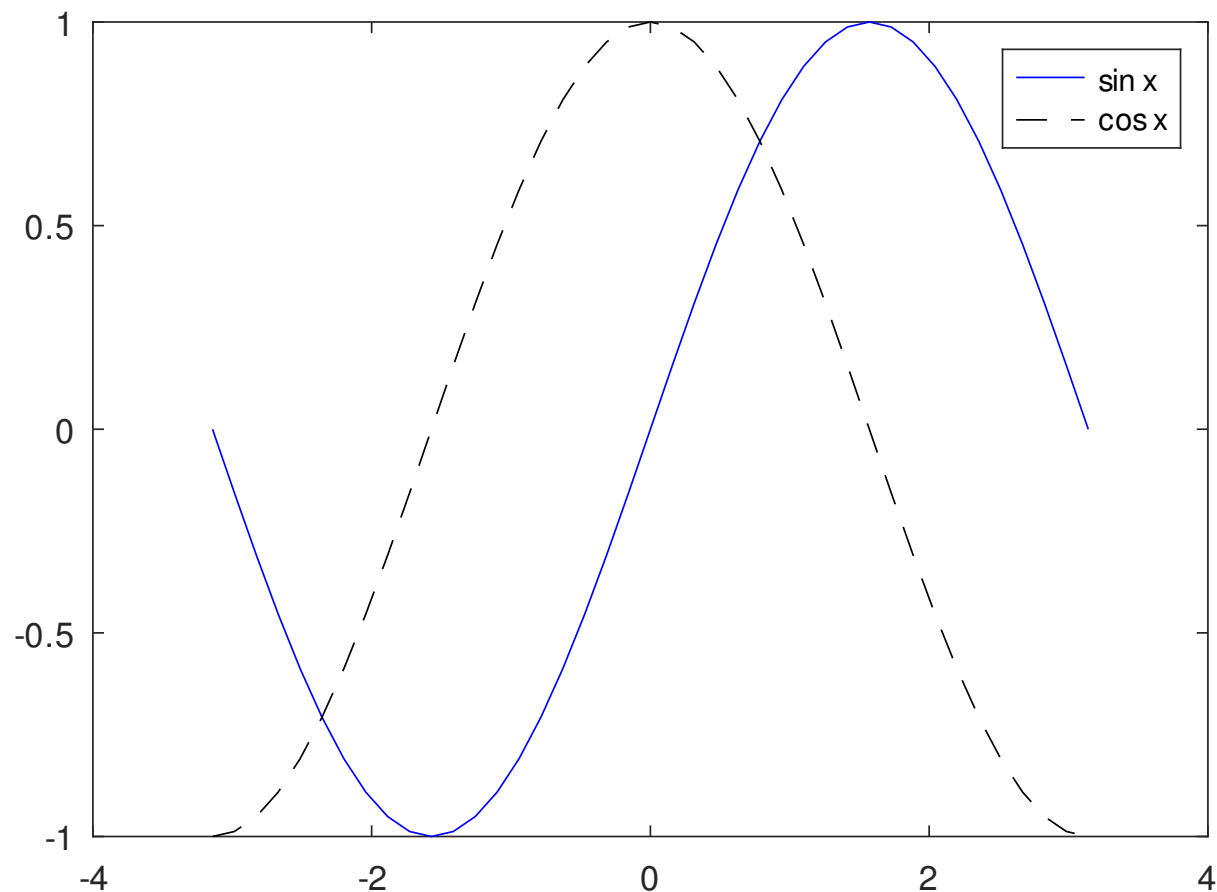
4 Plot Formatting Commands

- Basic plot command: plot(independent variable, dependent variable)
- Plot types: plot(), bar(), stem(), semilogy(), semilogx(), loglog()
- title(), xlabel(), ylabel(), xaxis(), yaxis(), legend()
- Line Style: Line width, Markers, colors
- subplot()
- grid on, hold on

title('string'): Adds the string as a title at the top of the plot.

xlabel('string'): Adds the string as a label to the x-axis.

ylabel('string'): Adds the string as a label to the y-axis.



axis([xmin xmax ymin ymax]) Sets the minimum and maximum limits of the x- and y-axes.

legend('string1','string2','string3') Creates a legend using the strings to label various curves (when several curves are in one plot). The location of the legend is specified by the mouse.

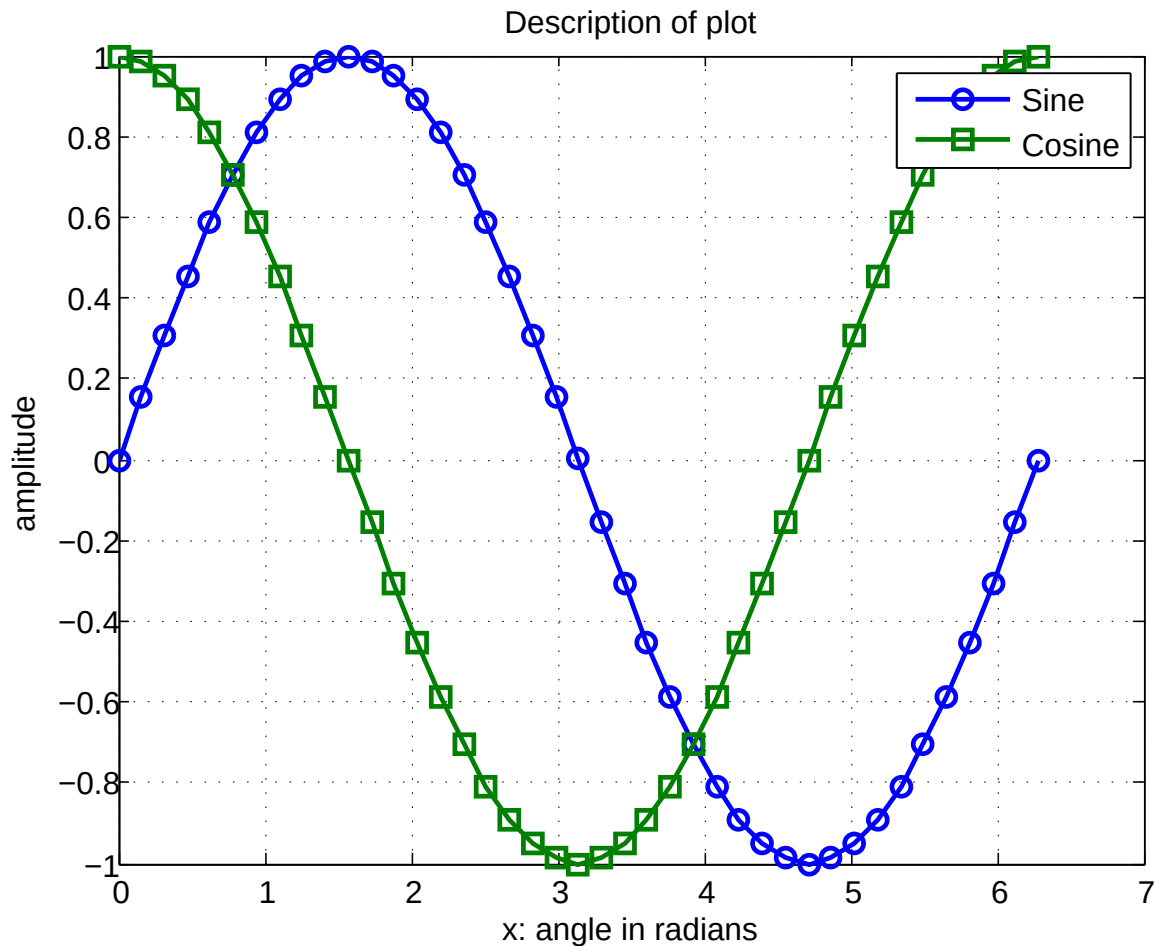
text(x,y,'string') Places the string (text) on the plot at coordinate x, y relative to the plot axes.

grid on To place grid on the plot area.

4.1 Example: Formatting Plots

```
clear all; close all; clc;
x=0:pi/20:2*pi
y1=sin(x);
y2=cos(x);
figure(1)
```

```
plot(x,y1,'-o',x,y2,'-s','LineWidth',1.5)
legend('Sine', 'Cosine')
xlabel('x: angle in radians')
ylabel('amplitude')
%axis([0.8 1.41 0 16])
title('Description of plot')
grid on
```



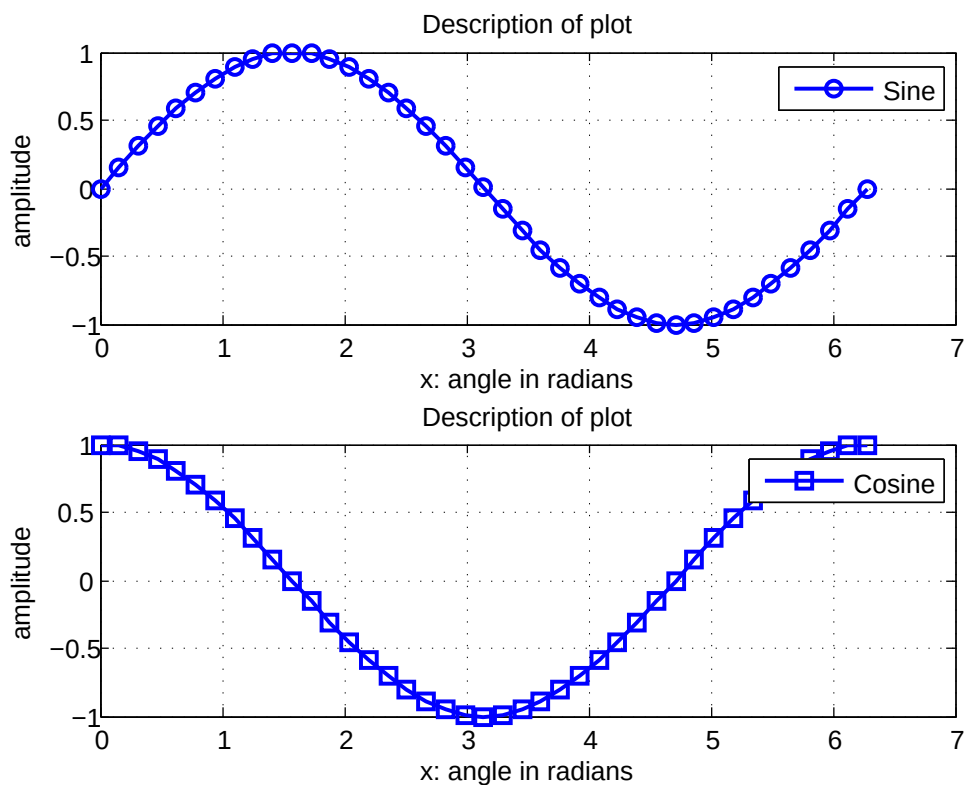
5 Subplot

Similarly we can plot more than one plots in side by side or one above the other using subplot() command.

Example: 2 x 1 subplots

```
subplot(2,1,1)
```

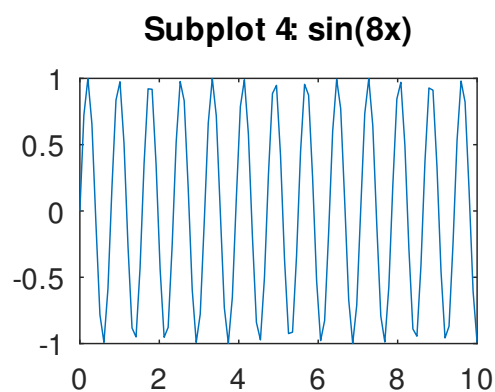
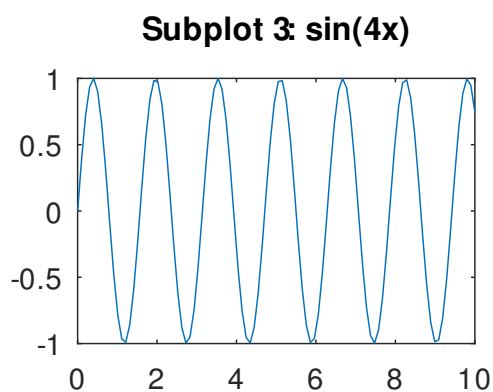
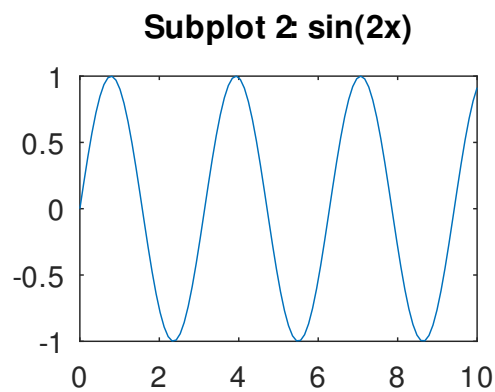
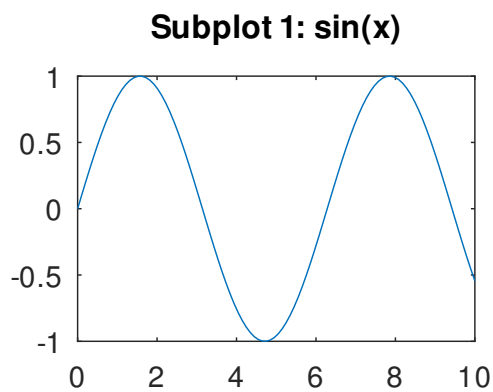
```
plot(x,y1,'-o','LineWidth',1.5)
legend('Sine')
xlabel('x: angle in radians')
ylabel('amplitude')
%axis([0.8 1.41 0 16])
title('Description of plot')
grid on
subplot(2,1,2)
plot(x,y2,'-s','LineWidth',1.5)
legend('Cosine')
xlabel('x: angle in radians')
ylabel('amplitude')
%axis([0.8 1.41 0 16])
title('Description of plot')
grid on
```



5.1 Example: 2 x 2 subplots

```
1 subplot(2,2,1)
2 x = linspace(0,10);
3 y1 = sin(x);
```

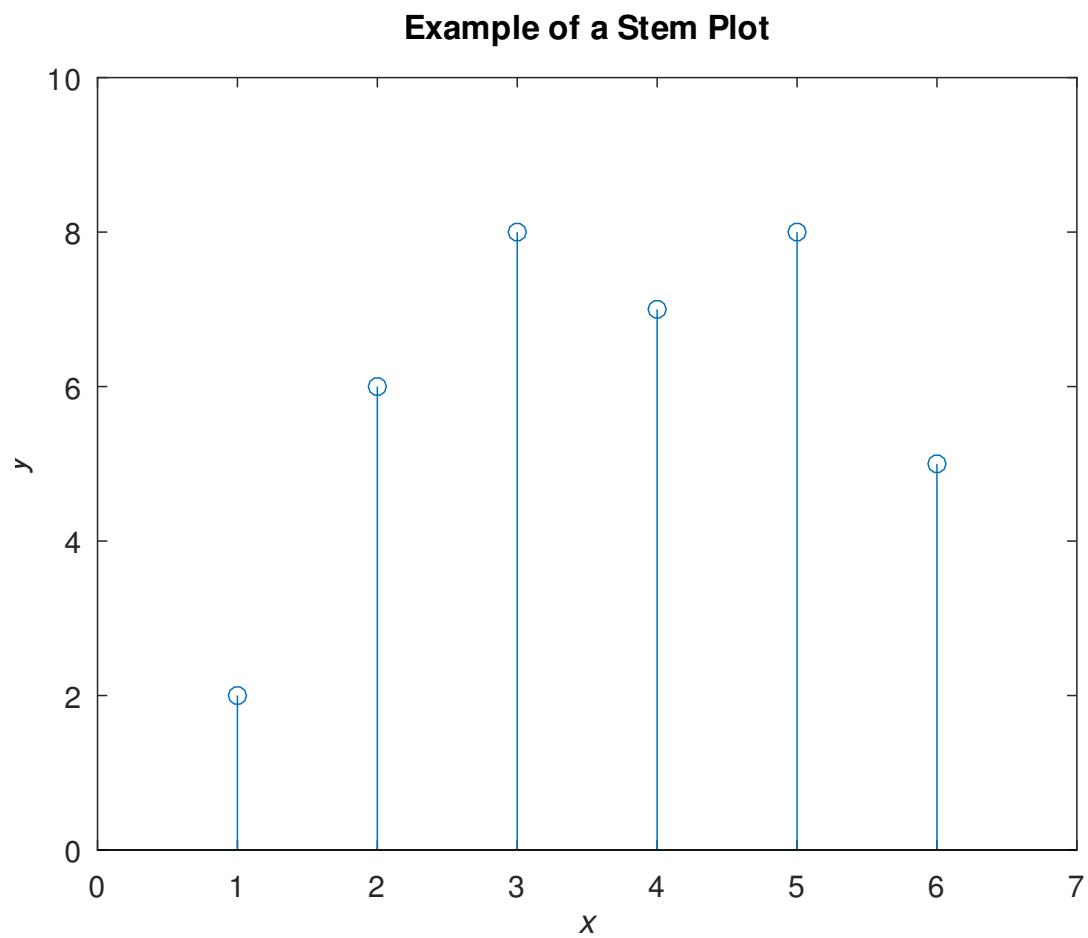
```
4 plot(x,y1)
5 title('Subplot 1: sin(x)')
6
7 subplot(2,2,2)
8 y2 = sin(2*x);
9 plot(x,y2)
10 title('Subplot 2: sin(2x)')
11
12 subplot(2,2,3)
13 y3 = sin(4*x);
14 plot(x,y3)
15 title('Subplot 3: sin(4x)')
16
17 subplot(2,2,4)
18 y4 = sin(8*x);
19 plot(x,y4)
20 title('Subplot 4: sin(8x)')
```



6 Stem Plot

It is used for discrete plots

```
1 x = [ 1 2 3 4 5 6];  
2 y = [ 2 6 8 7 8 5];  
3 stem(x,y);  
4 title('\bfExample of a Stem Plot');  
5 xlabel('\bf\itx');  
6 ylabel('\bf\ity');  
7 axis([0 7 0 10]);
```



7 Lab 02: Lab Tasks

1. Draw a 3-D mesh plot.
2. Plot the function $y(x) = e^{-0.5x} \sin 2x$ for 100 values of x between 0 and 10. Use a 2-point-wide solid blue line for this function. Then plot the function $y(x) = e^{-0.5x} \cos 2x$ on the same axes. Use a 3-point-wide dashed red line for this function. Be sure to include a legend, title, axis labels, and grid on the plots.
3. What is the difference between these two plots
 - `semilogx(x,y)`
 - `semilogy(x,y)`
4. What is the purpose of using `hold on` and `hold off` commands. Demonstrate the effect of using `hold on` and without using `hold on` with an example.